

Student Attendance Tracker - Mini Project Documentation

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1. Introduction

The Student Attendance Tracker is a Python-based application developed to manage and maintain attendance records of students. The system stores attendance information in a CSV file and allows users to mark attendance, view records, search for students, and generate attendance reports.

2. Objectives

- Automate attendance record management.
- Store attendance records in CSV format.
- Provide attendance statistics and reports.
- Demonstrate Python programming concepts.

3. Software Requirements

Python 3.x

Pandas Library

CSV Module

Thonny IDE

4. System Features

- Mark Attendance
- View Attendance Records
- Search Student
- Generate Attendance Report
- Store Data in CSV File

5. Working of the Project

The program creates an attendance.csv file if it does not exist. Users can enter roll number, student name, and attendance status. The information is stored in the CSV file. Pandas is used to read and display records, search students, and generate attendance statistics.

6. Concepts Used

- Classes and Objects
- Constructors (__init__)
- File Handling

- CSV Storage
- Pandas Data Analysis
- Loops and Conditions
- Functions

7. Sample Output

Attach screenshots of your program here before submission.

8. Conclusion

The Student Attendance Tracker successfully manages attendance records using Python. The project improves understanding of file handling, object-oriented programming, and data processing using Pandas.

9. Future Scope

- Add Date and Time Tracking
- Develop a GUI using Tkinter
- Integrate a Database
- Export Reports to Excel
- Add Student ID Validation